

Marketing Mix Modelling Platform

Auto-Modeller Bayesian MMM

Methodology & Technical Specification

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PART 1 — Executive Overview

1 Introduction

Marketing Mix Modelling (MMM) is a statistical approach used to quantify the impact of marketing activities on business outcomes such as sales, revenue, or prescriptions.

The MMM platform described in this document is an **automated Bayesian modelling system** designed to estimate the contribution and effectiveness of marketing investments using aggregated time-series data.

The system provides:

- marketing contribution estimates
- return on investment (ROI) measurement
- diminishing returns analysis
- response curves for optimization
- interpretable decomposition of business outcomes

The platform is designed to operate in an **automated modelling environment**, enabling consistent and scalable analysis across brands, markets, and time periods.

2 Marketing Mix Modelling Overview

Marketing activities influence consumer behavior across multiple touchpoints. These activities interact with external factors such as:

- economic conditions
- competitor activity
- seasonal demand
- pricing changes

MMM attempts to isolate the **incremental contribution of marketing activities** by statistically modelling the relationship between marketing exposures and business outcomes.

The key concept is:

$$Sales_t = Base_t + Marketing_t + ExternalFactors_t + Error_t$$

Where:

Component Description

Base	baseline demand
Marketing	impact of marketing
External	control variables
Error	unexplained variance

3 Why Bayesian MMM

Traditional MMM approaches rely on classical regression models. These approaches often suffer from instability when marketing variables are highly correlated.

The Bayesian approach offers several advantages.

Prior Knowledge

Bayesian models allow prior information about marketing effectiveness to be incorporated into the model.

Uncertainty Quantification

Bayesian models estimate full **posterior distributions** for each parameter.

This enables:

- ROI confidence intervals
 - uncertainty diagnostics
 - stability scoring
-

Improved Stability

Bayesian shrinkage reduces the risk of unrealistic coefficient estimates.

4 Platform Objectives

The Auto-Modeller platform was designed with the following objectives.

Automation

The modelling pipeline runs automatically with minimal manual intervention.

Scalability

The platform supports modelling at multiple granularity levels:

- national
 - region
 - region × brand
-

Interpretability

Model outputs are designed to be interpretable by business users.

Robustness

The platform includes:

- economic sanity checks
 - stability diagnostics
 - automated fallback mechanisms
-

5 Key Capabilities

The platform provides the following capabilities.

Marketing Attribution

Quantifies contribution of each marketing channel.

ROI Measurement

Estimates revenue return per unit of spend.

Effectiveness Analysis

Identifies channels that are over- or under-invested.

Response Curves

Estimates diminishing returns for each channel.

Automated Diagnostics

Ensures model reliability through statistical checks.

PART 2 — Data Architecture

6 Data Requirements

The MMM platform requires historical time-series data with a minimum observation window.

Recommended minimum duration:

Frequency Minimum Duration

Weekly 2–3 years

Monthly 3–5 years

7 Input Data Schema

All data must conform to a standardized schema.

date

region

sub_brand

variable

value

Example:

date	region	sub_brand	variable	value
2023-01-01	US	BrandA	clm_call	125
2023-01-01	US	BrandA	volume	4200

8 Variable Classification

Variables are defined in **info.csv**.

Each variable is assigned a type.

Type	Description
kpi_volume	model outcome
kpi_revenue	revenue metric
media	marketing exposure
control	external drivers

9 KPI Definition

The KPI represents the outcome being modelled.

Examples include:

- sales volume
- revenue
- prescriptions
- website conversions

When both **volume and revenue** exist:

volume → modelling KPI

revenue → ROI KPI

10 Media Variables

Media variables represent marketing exposures.

Examples include:

Channel	Example Metric
CLM calls	physician visits
Email campaigns	email sends
Webinars	attendee count

Channel	Example Metric
---------	----------------

Phone outreach	calls
----------------	-------

11 Control Variables

Control variables capture external factors affecting demand.

Examples include:

- competitor promotions
 - macroeconomic indicators
 - pricing changes
 - distribution changes
-

12 Spend Data Integration

Spend data is optional but required for ROI calculations.

Spend data structure:

```
date
region
sub_brand
media
spend
```

13 Data Validation Framework

The pipeline executes multiple validation steps.

Validation modules include:

```
validate_schema
validate_info
validate_data
cross_checks
```

14 Data Quality Rules

The following rules must be satisfied.

Time continuity

No missing dates.

Variable existence

All variables listed in **info.csv** must appear in data.

Non-negative exposures

Media exposures must be ≥ 0 .

KPI completeness

KPI values must not contain missing entries.

PART 3 — Data Transformation Layer

15. Data Transformation Overview

Before modelling begins, raw input data must undergo a series of transformations to ensure that it is suitable for statistical modelling.

The transformation layer serves several key purposes:

- standardizing time frequency
- aligning exposure and spend data
- handling missing values
- preparing features for modelling
- ensuring statistical stability

The transformation pipeline converts raw input datasets into **structured model tensors** used by the Bayesian model.

The transformation workflow is shown below.

Raw Data

↓

Validation

↓

Time Frequency Alignment

↓

Monthly → Weekly Conversion

↓

Exposure Scaling

↓

Feature Engineering

↓

Tensor Construction

Each step is described in the following sections.

16. Time Frequency Handling

Marketing and business datasets often contain observations recorded at different time frequencies.

Typical frequencies include:

Frequency Example

Daily digital campaigns

Weekly marketing exposures

Monthly sales reporting

To ensure consistency across variables, the MMM platform standardizes all datasets to a **single modelling frequency**.

The platform currently supports two frequencies:

weekly

monthly

However, the **default modelling frequency is weekly**.

Weekly frequency provides better statistical properties because:

- more observations improve estimation stability
- marketing effects are captured more accurately
- carryover dynamics are modeled more precisely

17. Monthly to Weekly Conversion

Many datasets are only available at monthly frequency.

However, modelling at monthly frequency often produces unreliable estimates due to the small number of observations.

For example:

Frequency Observations (3 years)

Monthly 36

Weekly 156

The platform therefore converts monthly data to weekly frequency when necessary.

17.1 Conversion Strategy

The monthly value is distributed across weeks contained within that month.

Let:

$$M_m$$

be the monthly value for month m .

If month m contains k weeks, then the weekly value is computed as:

$$W_t = \frac{M_m}{k}$$

for all weeks t within month m .

17.2 Preservation of Totals

The conversion procedure ensures that monthly totals remain unchanged.

For each variable:

$$\sum_{weeks} W_t = M_m$$

This reconciliation check guarantees that the conversion does not alter aggregate values.

17.3 Reconciliation Check

After conversion, the system verifies:

```
sum(original_monthly_values)
=
sum(converted_weekly_values)
```

If the difference exceeds a tolerance threshold, the pipeline raises an error.

18. Exposure Scaling

Marketing exposure variables can vary widely in magnitude.

Examples:

Channel	Typical Scale
---------	---------------

CLM Calls	0–100
-----------	-------

Emails	0–10,000
--------	----------

Impressions	millions
-------------	----------

Large differences in scale can cause instability in Bayesian estimation.

Therefore exposures are normalized before modelling.

18.1 Standardization

Exposure variables are standardized using z-score normalization.

For variable X :

$$X^{scaled} = \frac{X - \mu_X}{\sigma_X}$$

Where:

Parameter	Meaning
-----------	---------

μ_X	mean of variable
---------	------------------

σ_X	standard deviation
------------	--------------------

Standardization improves numerical stability during sampling.

18.2 KPI Scaling

The KPI variable is also standardized.

$$Y^{scaled} = \frac{Y - \mu_Y}{\sigma_Y}$$

After predictions are generated, values are converted back to the original scale.

19. Exposure Clipping

Extreme outliers in marketing exposures can distort model estimates.

To mitigate this risk, exposures are clipped to a reasonable range.

The clipping rule is:

$$X_{clipped} = clip(X, \epsilon, X_{max})$$

Where:

Parameter Meaning

ϵ small positive constant

X_{max} maximum allowed value

Typical values:

$\epsilon = 1e-6$

$X_{max} =$ large constant (e.g. $1e6$)

This prevents numerical issues during saturation transformations.

20. Spend Alignment

Marketing spend data is not used directly in the regression model but is required for ROI calculations.

Spend data must be aligned with exposure variables.

The alignment procedure includes:

1. matching media names between exposure and spend datasets
2. ensuring identical time frequencies
3. aggregating spend to the same geographic level

20.1 Spend Tensor

Spend values are stored in a tensor with the same structure as exposure data.

Tensor shape:

(T, G, SB, C)

Where:

Dimension Description

T	time periods
G	regions
SB	sub-brands
C	media channels

This alignment ensures that ROI calculations are consistent with exposure data.

21. Missing Value Handling

Missing values can arise due to:

- incomplete reporting
- delayed campaign reporting
- data ingestion errors

The transformation layer handles missing values using the following rules.

21.1 KPI Missing Values

Missing KPI observations are **not allowed**.

If KPI values are missing, the pipeline stops and raises an error.

21.2 Media Variables

Missing media exposures are replaced with zero.

This assumes that no activity occurred during that period.

21.3 Control Variables

Missing control variables are forward-filled or interpolated depending on variable type.

22. Feature Engineering

Several derived features are created during the transformation stage.

These features support downstream modelling.

22.1 Time Index

A numeric time index is generated.

$$t = 1, 2, 3, \dots, T$$

This variable is used for trend modelling.

22.2 Seasonal Components

Seasonal components are constructed using Fourier terms.

For each time period:

$$\begin{aligned} \sin\left(\frac{2\pi t}{52}\right) \\ \cos\left(\frac{2\pi t}{52}\right) \end{aligned}$$

These variables capture yearly seasonality.

23. Data Integrity Checks

After transformations, the system performs integrity checks.

These checks verify:

- tensor dimensions
 - variable consistency
 - absence of NaN values
 - correct scaling
-

23.1 Tensor Shape Validation

All tensors must satisfy the following structure:

$X_{\text{media}} \rightarrow (T, G, SB, C)$

$Y \rightarrow (T, G, SB)$

$X_{\text{control}} \rightarrow (T, G, SB, K)$

23.2 Frequency Verification

The system verifies that modelling frequency is weekly.

```
assert model_frequency == "weekly"
```

This ensures that all downstream modelling assumptions remain valid.

24. Tensor Construction

The final output of the transformation layer is a set of tensors used by the Bayesian model.

The primary tensors include:

Tensor	Description
--------	-------------

X_{media}	marketing exposures
--------------------	---------------------

X_{control}	control variables
----------------------	-------------------

Y	KPI variable
-----	--------------

X_{spend}	marketing spend
--------------------	-----------------

These tensors form the inputs to the probabilistic model.

25. Transformation Pipeline Summary

The data transformation pipeline performs the following steps:

1. data validation
2. frequency alignment
3. monthly to weekly conversion
4. exposure scaling

- 5. feature engineering
- 6. tensor construction

The output of this process is a structured dataset optimized for Bayesian estimation.

PART 4 — Model Architecture

20. Bayesian Regression Framework

The Auto-Modeller MMM platform is based on a **Bayesian hierarchical regression model** designed to estimate the impact of marketing exposures on business outcomes.

Let:

Symbol Meaning

- t time index
- g Region
- b sub-brand
- i marketing channel
- j control variable

The observed KPI is denoted as:

$$Y_{t,g,b}$$

The model decomposes the KPI into:

$$Y_{t,g,b} = Base_{t,g,b} + Media_{t,g,b} + Control_{t,g,b} + \epsilon_{t,g,b}$$

Where

Component Description

- Base underlying demand
- Media marketing contribution
- Control external drivers

Component Description

ε residual noise

21. Likelihood Function

The KPI is modeled using a **Gaussian likelihood**.

$$Y_{t,g,b} \sim \text{Normal}(\mu_{t,g,b}, \sigma)$$

Where

$$\mu_{t,g,b} = \text{Base}_{t,g,b} + \text{Media}_{t,g,b} + \text{Control}_{t,g,b}$$

The variance parameter:

$$\sigma \sim \text{HalfNormal}(1)$$

captures unexplained variation.

22. Baseline Component

Baseline demand represents the portion of KPI not driven by marketing.

Baseline includes three components:

$$\text{Base}_t = \text{Intercept} + \text{Trend}_t + \text{Seasonality}_t$$

22.1 Intercept

The intercept represents average demand level.

$$\begin{aligned}\alpha &\sim \text{Normal}(0,1) \\ \text{Intercept} &= \alpha\end{aligned}$$

22.2 Trend

Demand may increase or decrease over time due to market growth.

$$\text{Trend}_t = \beta_{\text{trend}} \cdot t$$

Prior:

$$\beta_{trend} \sim Normal(0,0.1)$$

23. Seasonality Modeling

Many industries exhibit seasonal demand patterns.

The platform models seasonality using **Fourier series**.

$$Season_t = \sum_{k=1}^K \left(a_k \sin\left(\frac{2\pi kt}{52}\right) + b_k \cos\left(\frac{2\pi kt}{52}\right) \right)$$

Where:

Parameter Meaning

K number of harmonics

Typically:

K = 2

This captures:

- yearly seasonality
- semi-annual seasonality

Priors:

$$a_k, b_k \sim Normal(0,0.1)$$

24. Control Variable Effects

External variables influence demand independently of marketing.

Control contribution:

$$Control_t = \sum_j \beta_j Z_{j,t}$$

Where:

Term Meaning

$Z_{j,t}$ control variable

β_j coefficient

Prior:

$$\beta_j \sim \text{Normal}(0,0.5)$$

PART 5 — Media Effect Modelling

25. Marketing Effect Transformation

Marketing exposures undergo two key transformations before entering the regression:

- 1 **Adstock transformation (carryover)**
- 2 **Saturation transformation (diminishing returns)**

This reflects real-world marketing dynamics.

26. Adstock (Carryover Effect)

Marketing exposure often has delayed impact.

Example:

- TV ad may influence consumers for several weeks
- email campaigns can generate follow-up engagement

To capture this effect, the platform uses an **exponential adstock function**.

26.1 Adstock Equation

For channel i :

$$\text{Adstock}_{i,t} = X_{i,t} + \theta_i \text{Adstock}_{i,t-1}$$

Where:

Parameter Meaning

$X_{i,t}$ exposure at time t

θ_i carryover coefficient

26.2 Adstock Interpretation

θ_i represents **retention rate**.

Example:

θ Interpretation

0 no carryover

0.3 moderate carryover

0.7 strong carryover

26.3 Adstock Prior

Retention parameters are bounded:

$$\theta_i \sim \text{Beta}(2,2)$$

This ensures:

$$0 < \theta < 1$$

27. Saturation (Diminishing Returns)

Marketing effectiveness decreases as exposure increases.

Example:

- first 100 TV impressions → strong impact
- next 100 impressions → smaller incremental effect

This phenomenon is modeled using a **Hill saturation function**.

27.1 Hill Function

$$S(X) = \frac{X}{\gamma + X}$$

Where:

Parameter Meaning

X adstocked exposure

γ saturation parameter

27.2 Saturation Behavior

γ Interpretation

small rapid saturation

large slower saturation

27.3 Saturation Prior

The platform uses a hierarchical prior:

$$\gamma_i \sim HalfNormal(\sigma)$$

Where:

$$\sigma \sim HalfNormal(1)$$

28. Media Contribution Equation

After transformations, media contribution is:

$$Media_t = \sum_i \beta_i \cdot S(Adstock_{i,t})$$

Where:

Parameter Meaning

β_i channel effectiveness

29. Media Coefficient Priors

Media coefficients are constrained to be positive.

$$\beta_i \sim HalfNormal(\sigma_\beta)$$

This reflects the assumption that marketing should not reduce sales.

30. Funnel Hierarchy Constraints

Marketing channels are categorized into funnel stages.

- Upper Funnel
- Mid Funnel
- Lower Funnel

The model applies ordering constraints:

$$\beta_{upper} \leq \beta_{mid} \leq \beta_{lower}$$

This reflects typical marketing performance patterns.

PART 6 — Bayesian Estimation

31. Posterior Distribution

The Bayesian framework estimates the posterior distribution:

$$P(\theta \mid data) = \frac{P(data \mid \theta)P(\theta)}{P(data)}$$

Where:

Term Meaning

($P(data \mid \theta)$)

Term	Meaning
------	---------

$P(\theta)$	prior
-------------	-------

$P(\theta \text{data})$	
---------------------------	--

32. Hamiltonian Monte Carlo

Posterior distributions are estimated using **Hamiltonian Monte Carlo (HMC)**.

The algorithm efficiently explores high-dimensional parameter spaces.

Implementation uses:

PyMC NUTS sampler

33. Sampling Configuration

The default configuration:

chains = 4

draws = 1000

tune = 1000

target_accept = 0.95

This configuration balances:

- sampling accuracy
- computational efficiency

34. Convergence Diagnostics

The platform evaluates convergence using:

Metric	Description
R-hat	convergence indicator
ESS	effective sample size
divergences	sampler failures

Acceptable thresholds:

R-hat < 1.05

ESS > 100

PART 7 — Contribution Decomposition

35. Delta Method Attribution

Channel contributions are estimated using the **delta attribution method**.

This approach measures incremental impact.

35.1 Full Prediction

$$\mu_{full} = Base + Media + Control$$

35.2 Channel Removal

For channel i :

$$\mu_{without,i} = Base + (Media - Contribution_i) + Control$$

35.3 Incremental Contribution

$$Contribution_i = \mu_{full} - \mu_{without,i}$$

36. Contribution Aggregation

Total contribution:

$$Contribution_i = \sum_t Contribution_{i,t}$$

PART 8 — ROI & Effectiveness

37. Revenue Conversion

If KPI is **volume**, contribution must be converted to revenue.

$$RevenuePerUnit = \frac{TotalRevenue}{TotalVolume}$$

Revenue contribution:

$$Contribution_{sales} = Contribution_{volume} \times RevenuePerUnit$$

38. ROI Calculation

Return on investment is defined as:

$$ROI_i = \frac{Contribution_{sales,i}}{Spend_i}$$

39. Effectiveness Metric

Effectiveness measures relative investment efficiency.

$$Effectiveness_i = \frac{ContributionShare_i}{SpendShare_i}$$

Where:

$$ContributionShare_i = \frac{Contribution_i}{TotalContribution}$$
$$SpendShare_i = \frac{Spend_i}{TotalSpend}$$

PART 9 — Response Curves

40. Response Curve Construction

Response curves simulate KPI outcomes under different spend levels.

For each channel:

spend_grid = 0.2 × spend → 2 × spend

41. Incremental Revenue

For each spend level:

$$Revenue(s) = ModelPrediction(s) - BaselinePrediction$$

42. Marginal ROI

Marginal ROI measures incremental revenue per additional unit of spend.

$$mROI = \frac{dRevenue}{dSpend}$$

This is approximated numerically using finite differences.

43. Diminishing Returns Point

The **diminishing returns threshold** is defined where:

$$mROI < ROI_{threshold}$$

Typical threshold:

1.2

44. Optimal Economic Point

Optimal investment occurs where:

$$mROI = 1$$

This indicates the break-even point for additional investment.

45. ROI Stability

ROI uncertainty is quantified using posterior intervals.

$$StabilityRatio = \frac{ROI_{p95} - ROI_{p05}}{|ROI_{mean}|}$$

Classification:

Ratio Stability

<1 stable

1-2 moderate

2 | unstable |

PART 10 — Response Curves

Response curves represent the relationship between **marketing investment and incremental business outcomes**.

These curves enable marketers to understand:

- how marketing performance changes with spend
- where diminishing returns occur
- where optimal investment levels lie

Response curves form the basis for **budget allocation decisions** and future optimization systems.

46. Marginal ROI Curves

Marginal ROI (mROI) measures the **incremental return generated by an additional unit of marketing investment**.

While standard ROI measures the return from total spend, marginal ROI measures the return from **the next incremental investment**.

46.1 Definition

Let:

$$Revenue(s)$$

represent predicted revenue when spend equals s .

Marginal ROI is defined as the derivative:

$$mROI(s) = \frac{dRevenue(s)}{ds}$$

46.2 Numerical Approximation

Because the model does not have a closed-form derivative, marginal ROI is computed using finite differences.

Let:

$$s_1, s_2$$

be two adjacent spend levels.

$$mROI(s) = \frac{Revenue(s_2) - Revenue(s_1)}{s_2 - s_1}$$

This approach approximates the slope of the revenue curve.

46.3 Spend Grid Construction

To compute marginal ROI, the system simulates multiple spend levels.

For each marketing channel i :

$$SpendGrid = [0.2 \times Spend_i, \dots, 2.0 \times Spend_i]$$

Typically:

40 grid points

are generated across this range.

46.4 Revenue Simulation

For each spend level:

1. marketing exposure is scaled
2. the model recomputes predictions
3. incremental revenue is calculated

The resulting curve represents the relationship between **spend and incremental revenue**.

47. Spend Response Functions

Spend response functions describe how marketing investment affects business outcomes.

These functions typically exhibit **non-linear behavior**.

47.1 General Shape

Marketing response curves usually follow an **S-shaped pattern**.

Low spend → under-investment

Mid spend → efficient growth

High spend → diminishing returns

This pattern emerges naturally from the combination of:

- adstock carryover
 - saturation effects
-

47.2 Mathematical Form

The response function is derived from the transformed exposure variable:

$$Revenue(s) = \sum_i \beta_i \cdot S(Adstock(X_i(s)))$$

Where:

Component Meaning

$X_i(s)$	exposure as a function of spend
Adstock	carryover transformation
S	saturation function

47.3 Exposure Scaling

Exposure levels are assumed to scale proportionally with spend:

$$X_i(s) = X_i \cdot \frac{s}{Spend_i}$$

This assumption allows the model to simulate alternative investment levels.

48. Diminishing Returns Detection

One of the most valuable insights produced by MMM is identifying **the point at which additional investment becomes inefficient**.

This point is referred to as the **diminishing returns threshold**.

48.1 Definition

Diminishing returns occur when:

$$mROI < ROI_{threshold}$$

A commonly used threshold is:

$$ROI = 1.2$$

Meaning that every additional dollar spent returns only \$1.20 or less.

48.2 Interpretation

When marginal ROI falls below this threshold:

- additional investment becomes inefficient
 - budget should be reallocated to higher-performing channels
-

48.3 Practical Example

Spend Revenue mROI

\$100K	\$200K	2.0
\$200K	\$320K	1.2
\$300K	\$350K	0.5

The diminishing returns point lies near **\$200K**.

49. Key ROI Points

For each marketing channel, the platform extracts three key investment points.

49.1 Current Investment Point

Represents the current marketing spend.

Metrics reported:

- spend
 - exposure
 - predicted revenue
 - ROI
 - marginal ROI
-

49.2 Diminishing Returns Point

This point represents the level at which marginal ROI begins to decline significantly.

Formally:

$$mROI(s) < ROI_{threshold}$$

This indicates that additional investment produces diminishing gains.

49.3 Optimal Economic Point

The optimal economic point is defined as the investment level where:

$$mROI(s) = 1$$

At this point:

- each additional dollar spent generates one dollar in revenue
 - investments beyond this point may reduce profitability
-

PART 11 — Model Diagnostics

Model diagnostics ensure that the MMM results are **statistically valid and economically meaningful**.

Diagnostics are performed automatically after model estimation.

50. Bayesian Diagnostics

Bayesian diagnostics evaluate the quality of the posterior sampling process.

The platform evaluates three key indicators.

50.1 R-hat Statistic

R-hat measures the convergence of Markov chains.

$$\hat{R} = \frac{\text{Variance}_{\text{between}}}{\text{Variance}_{\text{within}}}$$

Interpretation:

R-hat Meaning

≈ 1 converged

1.05 | potential non-convergence |

50.2 Effective Sample Size

Effective sample size (ESS) measures the number of independent samples produced by the sampler.

Low ESS indicates that samples are highly correlated.

Recommended threshold:

ESS > 100

50.3 Divergences

Divergences occur when the sampler fails to explore the posterior properly.

A small number of divergences is acceptable.

However:

divergences > 100

may indicate model instability.

51. Fit Metrics

Fit metrics evaluate how well the model predicts observed data.

51.1 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{N} \sum (y - \hat{y})^2}$$

Measures absolute prediction error.

51.2 Mean Absolute Percentage Error

$$MAPE = \frac{1}{N} \sum \left| \frac{y - \hat{y}}{y} \right|$$

Measures relative prediction error.

51.3 R-squared

$$R^2 = 1 - \frac{SS_{residual}}{SS_{total}}$$

Measures the proportion of variance explained by the model.

51.4 Bayesian R²

Bayesian R² estimates model fit across posterior samples.

$$R^2 = \frac{Var(\mu)}{Var(\mu) + Var(\epsilon)}$$

Where:

Term Meaning

μ predicted values

Term Meaning

ϵ residual variance

52. Economic Sanity Checks

Statistical models can sometimes produce results that are mathematically valid but economically implausible.

The platform includes automated economic checks.

52.1 Excessive ROI Detection

Channels with extremely high ROI are flagged.

Example threshold:

$ROI > 50$

This may indicate:

- low spend levels
 - insufficient variation
 - model overfitting
-

52.2 Contribution Consistency

Total media contribution must not exceed the KPI.

The system verifies:

$$\sum_i Contribution_i \leq Y$$

If violations occur, reconciliation adjustments are applied.

52.3 Spend-Contribution Imbalance

Channels with extremely low spend but high contribution are flagged.

These cases require further investigation.

PART 12 — Platform Outputs

The MMM platform generates a standardized set of outputs used for analysis and visualization.

These outputs serve three main purposes:

- modelling diagnostics
- marketing performance measurement
- dashboard visualization

53. Output Dataset Definitions

The primary output datasets include the following.

53.1 ROI Dataset

File:

roi.csv

Contains aggregated performance metrics for each channel.

Columns include:

Column	Description
media	marketing channel
total_spend	total investment
total_contribution_volume	incremental KPI
total_contribution_sales	incremental revenue
roi_mean	average ROI
roi_p05	lower confidence interval
roi_p95	upper confidence interval
effectiveness	efficiency score
roi_stability	uncertainty classification

53.2 Media Contribution Dataset

File:

media_contribution.csv

Provides time-series contribution estimates.

Column	Description
date	observation date
region	geographic unit
sub_brand	brand
media	channel
contribution	incremental KPI

53.3 Base Contribution Dataset

File:

base_contribution.csv

Contains the baseline component of the KPI.

53.4 Response Curve Dataset

File:

marginal_roi_curves.csv

Contains simulated spend levels and predicted returns.

Columns include:

Column	Description
media	channel
spend	simulated spend
roi	predicted ROI
contribution	incremental contribution

53.5 ROI Key Points

File:

roi_key_points.csv

Contains important investment thresholds.

Column	Description
--------	-------------

point_type	current / optimal / diminishing
------------	---------------------------------

spend	investment level
-------	------------------

roi	ROI
-----	-----

mroi	marginal ROI
------	--------------

54. Visualization Data Structures

Outputs are designed to support dashboard visualization.

Typical visualizations include:

- contribution waterfall charts
 - ROI comparison charts
 - marginal ROI curves
 - spend allocation dashboards
-

55. Dashboard Integration

The output datasets can be integrated into BI tools such as:

Tool	Usage
------	-------

Power BI	executive dashboards
----------	----------------------

Tableau	marketing performance analysis
---------	--------------------------------

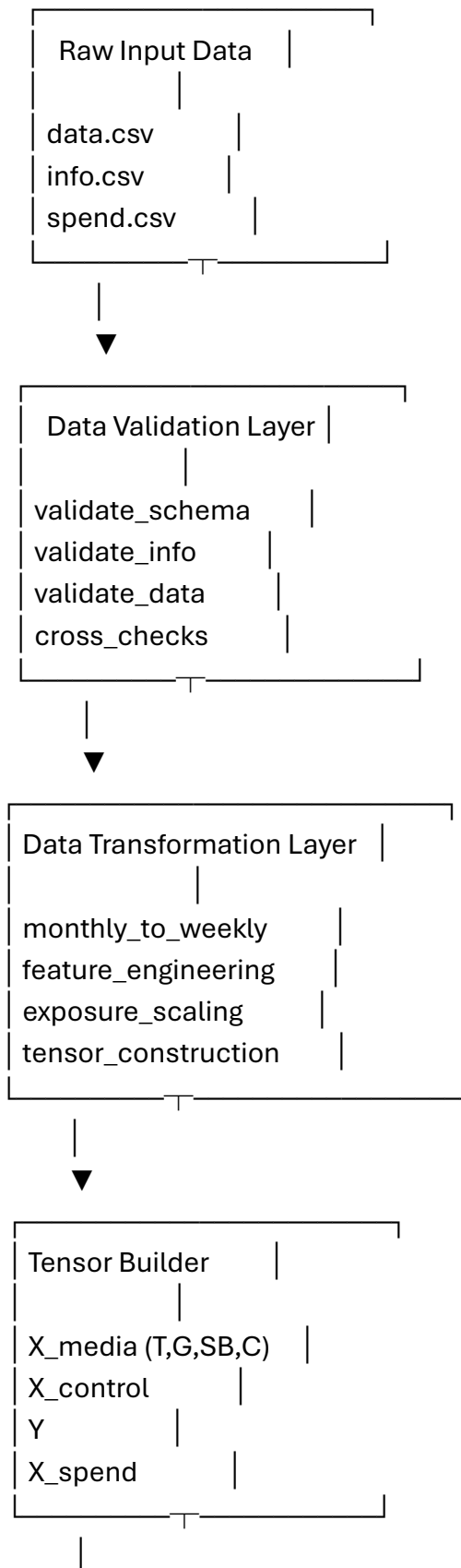
Custom UI	SaaS product interface
-----------	------------------------

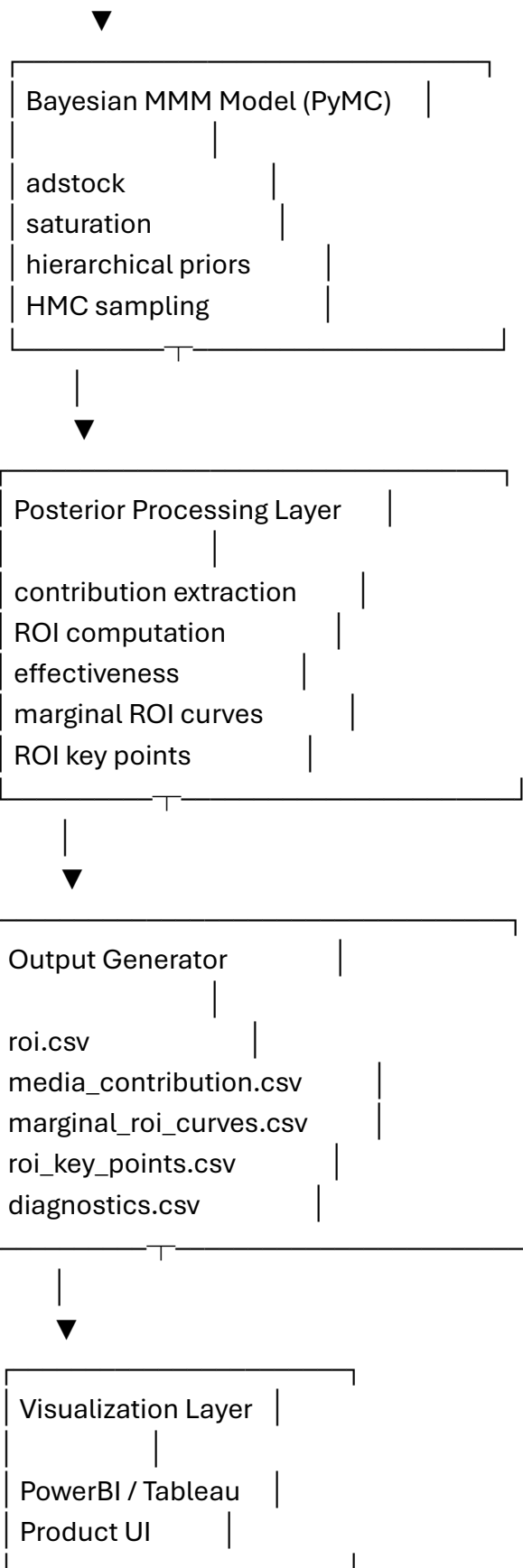
The standardized CSV structure enables seamless integration with visualization platforms.

PART 13 — System Architecture

56. MMM Pipeline Architecture

This diagram shows the **end-to-end modelling pipeline** from data ingestion to outputs.





57. Code Module Architecture

This diagram shows how the **codebase is organized**.

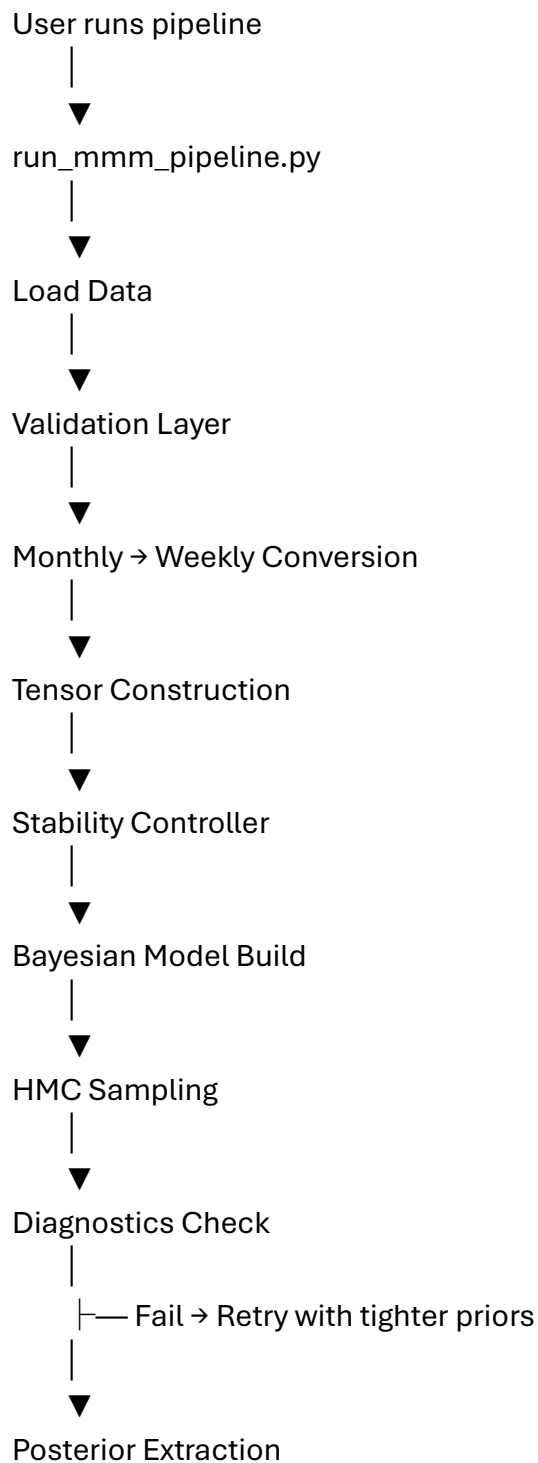
```
src/
|
|— io/
|   |— load_data.py
|   |— load_spend.py
|   └— write_outputs.py
|
|— validation/
|   |— validate_info.py
|   |— validate_data.py
|   |— cross_checks.py
|   └— small_data_checks.py
|
|— preprocessing/
|   └— monthly_to_weekly.py
|
|— parser/
|   └— build_tensors.py
|
|— model/
|   |— pymc_mmm.py
|   |— sample_model.py
|   |— stability_controller.py
|   |— contributions.py
|   |— roi.py
|   |— effectiveness.py
|   |— marginal_roi.py
|   |— roi_key_points.py
|   └— economic_sanity_checks.py
|
|— diagnostics/
|   └— model_diagnostics.py
|
└— run_mmm_pipeline.py
```

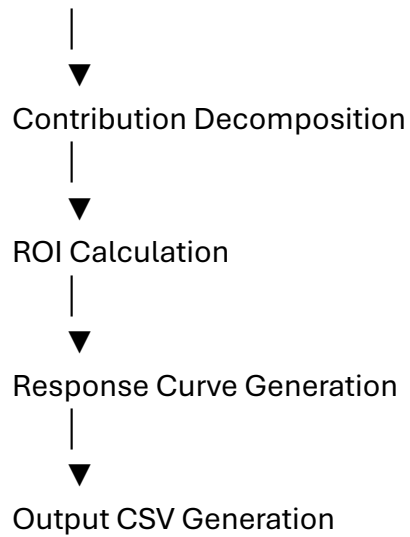
This modular architecture ensures:

- separation of concerns
 - easier testing
 - easier maintenance
-

58. Model Execution Flow

This diagram explains **what happens when the model runs**.



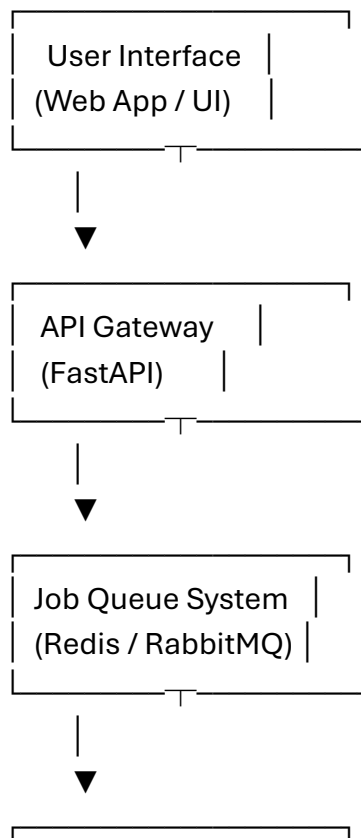


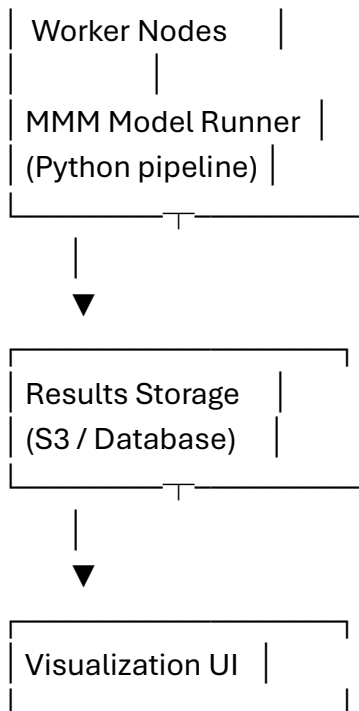
PART 14 — Deployment Architecture

This is the **important section for turning the MMM platform into a product.**

59. Batch Execution Architecture

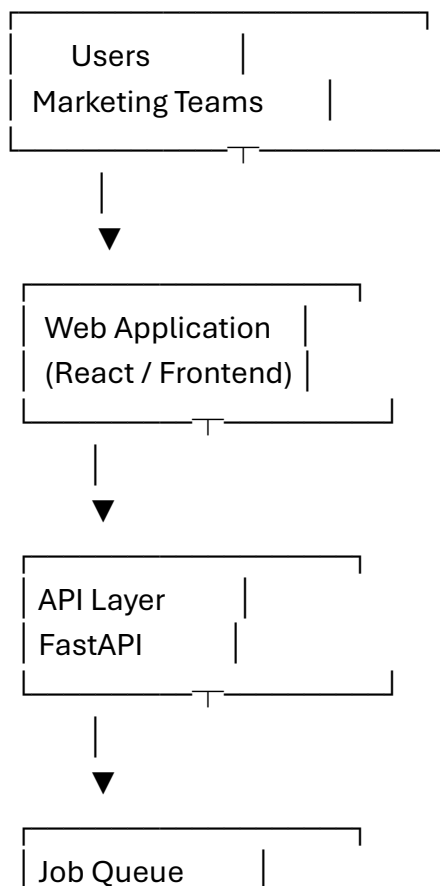
MMM models take **several minutes to run**, so they should run as **background jobs**, not synchronous APIs.

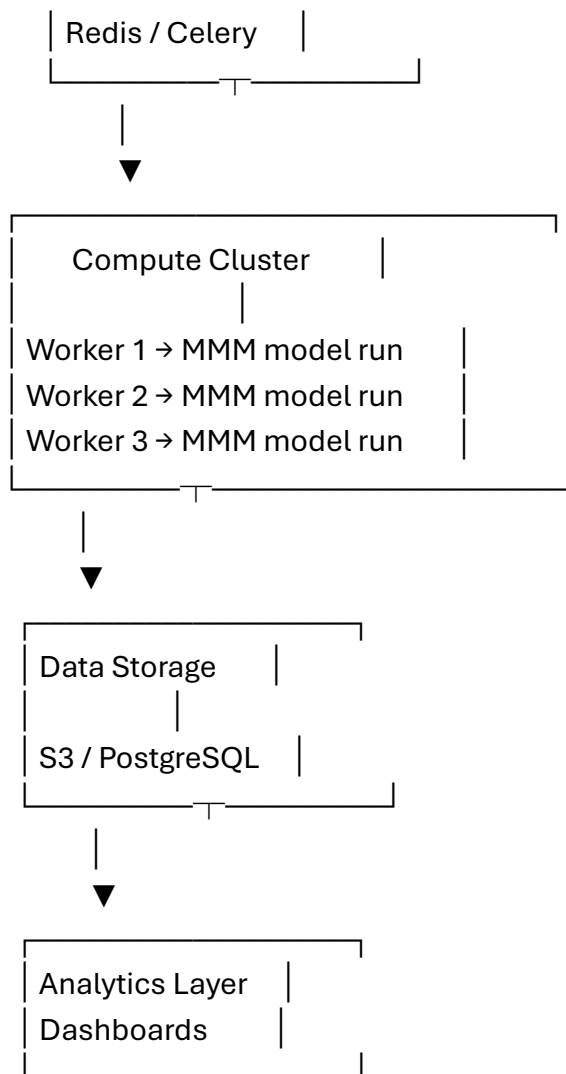




60. SaaS Deployment Architecture

For a **multi-client MMM product**, architecture typically looks like this.





61. Model Execution Time

Typical runtime:

Model Type	Runtime
National MMM	3–5 minutes
Region MMM	8–12 minutes
Region × Brand MMM	15–25 minutes

Therefore the platform uses **asynchronous batch processing**.

62. Scalability Strategy

Scaling is achieved through **horizontal worker scaling**.

More MMM jobs → add more workers

Example:

Workers Models/hour

1	12
5	60
10	120